Loop and Node Methods for Linear Networks

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The analysis of a network begins with the elements it contains and the way they are joined. Each branch carries a current, and each node has a potential measured against a common reference. Kirchhoff's laws relate these quantities: the currents entering a node sum to zero, and the voltages around any closed loop sum to zero. With Ohm's law for every resistor, these relations give as many independent equations as there are unknowns, so the circuit can be solved

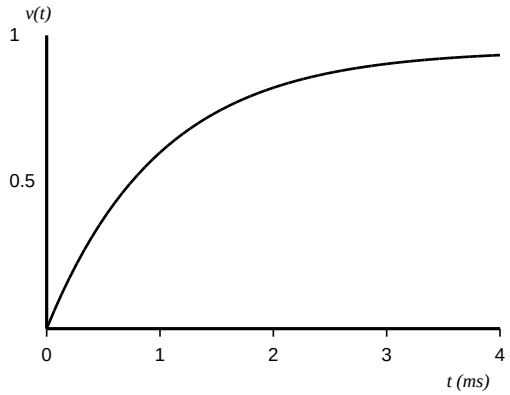

Fig. 1. Step response of the RC network.

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Fig. 2. Signal path of the measurement.

TABLE I Element values

Element	Value	Tolerance
R1	4.7 k	1 %
R2	10 k	1 %
C1	100 n	5 %

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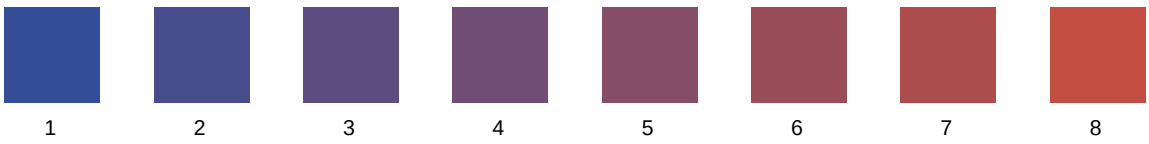

Fig. 3. The eight test configurations, left to right.

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